

Lukas Wölfl

Data & AI Engineer | AWS Certified Solutions Architect | IIoT & Energy

Freiburg, Baden-Württemberg, Germany

Executive Profile

An engineering-driven professional bridging data platforms, cloud architecture, and industrial intelligence. Specializes in building production-grade, scalable data pipelines, agentic AI/MLOps workflows, and automated infrastructure (IaC/CI-CD), with deep domain expertise in industrial IoT (IIoT) and energy systems.

Certifications & Education

Certifications

- **AWS Certified Solutions Architect – Associate** | Amazon Web Services (Issued: May 2026 · Expires: May 2029)
- **LangChain: Develop LLM powered applications** | Udemy (Issued: Mar 2025)
- **Statistik & Mathematik für Data Science & Data Analytics** | Udemy (Issued: May 2022)
- **Machine Learning von A-Z: Python & R** | Udemy (Issued: Jun 2020)
- **Python Bootcamp: Vom Anfänger zum Profi, inkl. Data Science** | Udemy (Issued: Jun 2020)

Education

- **M.Sc. in Industrial Engineering and Management (Electrical Engineering)**
Paderborn University | 2017 – 2020
 - **Focus:** Intelligent electricity grids, power electronics, data science, and Business Analytics/ML.
 - **Master’s Thesis:** Developed ML models to detect bearing damage in drive axles (Grade: 1.0).
- **Advanced Power Systems Research Exchange**
University of the Witwatersrand | 2018
 - **Focus:** DAAD-funded research on Microgrids, decentralized energy systems, and controller design.
- **B.Sc. in Industrial Engineering and Management (Electrical Engineering)**
Paderborn University | 2012 – 2017
 - **Focus:** Signal theory, automation technology, corporate strategy, and industrial production.

Technical Skills Matrix

Category	Skills & Technologies

Data & AI Engineering	Python, SQL, Google BigQuery, Reverse-ETL, LLMs, LLMOps, LangChain, Predictive Maintenance, Condition Monitoring, MLOps, Agentic AI Workflows
Cloud & DevOps	Amazon Web Services (AWS), Google Cloud Platform (GCP), Terraform, Docker, Infrastructure as Code (IaC), CI/CD
IIoT & Energy Systems	MQTT, E-Mobility, Charging Infrastructure, Storage Modeling, PV Yield Simulation, Smart Cities, Power Distribution
Tools & Legacy Systems	Git, GitLab, SAP ERP, MS Project, Visio, VBA / Excel Automation, MATLAB

Professional Experience

Ingersoll Rand

Data Scientist (Full-Time) | Oct 2022 – Present | Bottrop, Germany (Hybrid)

- **Data Pipeline Engineering:** Developed modular Reverse-ETL pipelines using Python and BigQuery, implementing automated scheduling via cron jobs.
- **Data Tools & Software Engineering:** Built centralized internal Python libraries to standardize data integration and visualization workflows across teams.
- **Automated Insights:** Designed SQL-based ELT pipelines to pinpoint commercial opportunities, automating direct lead generation pushes into Salesforce.
- **Analytics Automation:** Engineered a configuration-driven reporting system for KPIs, handling automated PDF/PPT generation and SaaS deployment.
- **Predictive Maintenance:** Deployed Python-based machine learning models for equipment condition monitoring, identifying physical degradation.
- **Agentic AI:** Integrated functional AI agents to automate exploratory data analysis (EDA), code documentation, product epics, and Jira administration.
- **Innovation & Leadership:** Led complex engineering projects from concept to production, mentored junior developers, and successfully advanced IIoT patent submissions to the corporate council.

Dortmunder Energie- und Wasserversorgung GmbH (DEW21)

Total Duration: 2 yrs 2 mos | Dortmund, Germany (Hybrid)

- **Engineer for Charging Infrastructure E-Mobility** | Jan 2022 – Sep 2022

- **Process Automation:** Developed a Python-based load profile analysis application; built automated VBA engines to generate planning documentation.
- **Network Strategy:** Conducted competitive benchmarking and created complex economic models to determine the feasibility of public charging networks.
- **Project Management:** Managed infrastructure rollouts from technical conceptualization through execution utilizing MS Project and Visio.
- **Trainee in E-Mobility | Aug 2020 – Dec 2021**
 - **Technical Consulting:** Developed tailored power supply and charging solutions for commercial B2B clients, municipal authorities, and public housing associations.
 - **Operations & Compliance:** Oversaw on-site technical construction phases and ran comprehensive quality checks for grid compliance.

Lenze

Total Duration: 1 yr 8 mos | Aerzen / Extertal, Germany (On-site)

- **Master's Candidate - Innovation | Apr 2019 – Feb 2020**
 - **Signal Processing & ML:** Formulated frequency-domain feature extraction methods to isolate structural drive-axle damage using statistical machine learning.
 - **Systems Engineering:** Commissioned IoT communication systems for real-time data acquisition and parameterized servo frequency converters.
- **Working Student – Product Management Service | Nov 2017 – Jul 2018**
 - Supported asset management optimization through hardware/software compatibility testing using MQTT protocols.

ENTEGRO Photovoltaik-Systeme GmbH

Working Student | Jun 2016 – Sep 2017 | Fröndenberg, Germany (On-site)

- **Energy Modeling:** Built solar storage simulation software in Excel/VBA utilizing real-world loading data to execute yield forecasting.
- **Electrical Compliance:** Physically installed and commissioned utility-scale PV systems under strict DIN VDE 0100-600 testing parameters.

TRILUX Deutschland

Total Duration: 6 yrs 5 mos | Arnsberg, Germany (On-site)

- **Bachelor Candidate - Outdoor Lighting Sales | Oct 2015 – Mar 2016**
 - Researched the economic viability of smart, sensor-equipped IoT lighting networks for urban parking spot detection.
- **Working Student – Special Lighting Team | Oct 2012 – Sep 2015**
 - Aided electrical hardware development via circuit diagram creation, component selection, and thermal analysis using infrared thermography.
- **Trainee – Electronics Technician for Industrial Engineering | Aug 2008 – Jun 2011**

- Executed industrial machinery troubleshooting, control cabinet wiring, and basic PLC/sensor parameterization.